

INDUSTRIAL SAFETY MONITORING AND CONTROL SYSTEM

Using Arduino UNO Microcontroller

Simulated in SimulIDE | Programmed in Arduino IDE

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1. Introduction

Industrial safety is one of the most critical aspects of modern manufacturing and production environments. The detection of fire, gas leaks, and abnormal temperature conditions can prevent major accidents, loss of life, and property damage. This project presents a simulated Industrial Safety Monitoring and Control System built using an Arduino UNO microcontroller, designed and tested in SimulIDE simulation software.

The system continuously monitors environmental conditions and triggers visual and audio alarms when dangerous levels are detected. A manual reset button allows operators to acknowledge and clear alarms after the situation is resolved.

2. Objectives

- To design and simulate an industrial safety monitoring system using Arduino UNO.
- To detect dangerous temperature and gas levels using analog sensors.
- To provide visual warnings using LED indicators when danger is detected.
- To provide audio alarm using a buzzer when danger threshold is exceeded.
- To implement a manual alarm reset button for operator control.
- To demonstrate the working of an embedded safety system in SimulIDE.

3. Components Used

The following hardware and software components were used in this project:

Component	Qty	Purpose	Arduino Pin
Arduino UNO	1	Main microcontroller unit	—
Potentiometer (1kΩ)	2	Simulates Temperature & Gas sensors	A0, A1
LED (Red)	2	Visual danger warning indicators	Pin 8, Pin 9
Buzzer (Audio Out)	1	Audio alarm on danger detection	Pin 10
Push Button	1	Manual alarm reset by operator	Pin 12
SimulIDE Software	—	Circuit simulation environment	—
Arduino IDE	—	Code writing and HEX generation	—

4. Circuit Diagram

The following figure shows the complete circuit designed and simulated in SimulIDE. All components are connected to the Arduino UNO as per the pin configuration described in this report.

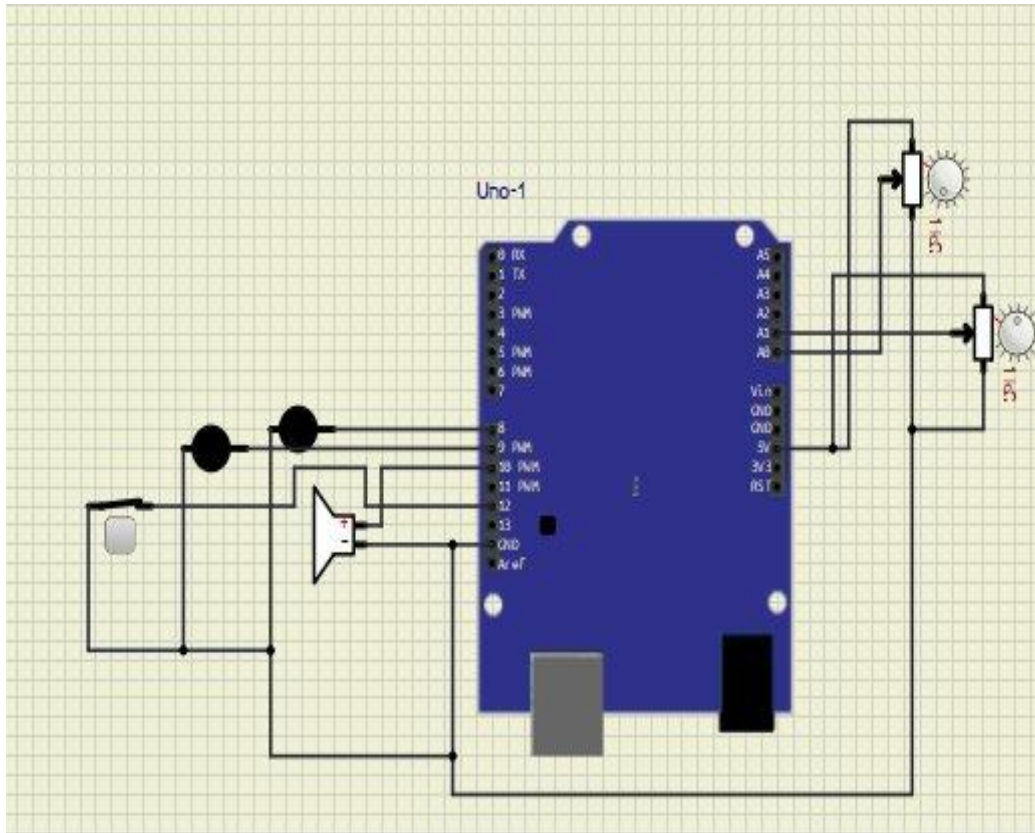


Figure 1: Industrial Safety Monitoring & Control Circuit (SimulIDE)

5. Pin Configuration Summary

Signal	Arduino Pin	Direction	Description
Temperature Sensor	A0	INPUT	Analog read from Potentiometer 1
Gas Sensor	A1	INPUT	Analog read from Potentiometer 2
Temperature LED	Pin 8	OUTPUT	Turns ON when temp > threshold
Gas LED	Pin 9	OUTPUT	Turns ON when gas > threshold
Buzzer	Pin 10	OUTPUT	Sounds alarm when danger detected
Reset Button	Pin 12	INPUT	Clears alarm when pressed

6. Circuit Description

The circuit was designed and simulated in SimulIDE. Each component is connected to the Arduino UNO as described below:

6.1 Temperature Sensor Simulation

A potentiometer (1k Ω) simulates the LM35 temperature sensor. The wiper (middle) pin connects to analog pin A0. The top pin connects to 5V and the bottom pin to GND. Rotating the knob changes the analog value, simulating temperature rise or fall.

6.2 Gas Sensor Simulation

A second potentiometer (1k Ω) simulates the MQ-2 gas sensor connected to analog pin A1. Increasing the knob value simulates rising gas concentration in the industrial environment.

6.3 LED Warning Indicators

Two red LEDs connect to digital pins 8 and 9. LED 1 (Pin 8) is the temperature warning and LED 2 (Pin 9) is the gas warning. Both LEDs share a common GND connection and turn ON independently based on their respective sensor readings.

6.4 Buzzer Alarm

An Audio Out component (buzzer) connects to digital pin 10. The buzzer activates only when danger is detected and is completely silent during normal safe operation. The Buzzer mode is enabled in the component properties within SimulIDE.

6.5 Reset Button

A push button on digital pin 12 uses INPUT_PULLUP configuration. When pressed, it resets the alarm state and turns OFF all LEDs and the buzzer, simulating a real-world alarm acknowledgement system used in industrial control panels.

6.6 Common Ground

All component GND pins share a single common ground wire. This is standard professional practice in electronic circuit design, ensuring consistent voltage reference and simplifying wiring.

7. Arduino Source Code

The following code was written in Arduino IDE and uploaded to the Arduino UNO via a HEX file in SimulIDE. The code reads sensor values every 500ms and controls the LEDs and buzzer based on threshold comparisons.

```
// =====  
//   Industrial Safety Monitoring & Control System  
//   Arduino UNO  
//   Team: Almas(006), Malaika(057), Maryam(061), Tehreem(089)  
// =====  
  
// --- Pin Definitions ---  
#define TEMP_SENSOR  A0    // Potentiometer 1 - Temperature  
#define GAS_SENSOR   A1    // Potentiometer 2 - Gas/Smoke  
#define LED_TEMP      8     // Temperature Warning LED  
#define LED_GAS       9     // Gas Warning LED  
#define BUZZER        10    // Alarm Buzzer  
#define RESET_BTN     12    // Reset Button  
  
// --- Threshold Values (0 - 1023) ---  
#define TEMP_THRESHOLD 600  
#define GAS_THRESHOLD  600  
  
// --- Variables ---  
bool alarmActive = false;  
  
void setup() {  
    pinMode(LED_TEMP, OUTPUT);  
    pinMode(LED_GAS,  OUTPUT);  
    pinMode(BUZZER,   OUTPUT);  
    pinMode(RESET_BTN, INPUT_PULLUP);  
  
    // Ensure everything OFF at startup  
    digitalWrite(BUZZER, LOW);  
    digitalWrite(LED_TEMP, LOW);  
    digitalWrite(LED_GAS, LOW);  
  
    Serial.begin(9600);  
    Serial.println("=====");  
    Serial.println("  Industrial Safety System ON  ");  
    Serial.println("=====");  
}  
  
void loop() {  
    int tempValue = analogRead(TEMP_SENSOR);  
    int gasValue  = analogRead(GAS_SENSOR);  
  
    Serial.print("Temp Value: ");  
    Serial.print(tempValue);  
    Serial.print(" | Gas Value: ");  
    Serial.println(gasValue);  
  
    // Reset alarm flag at start of each loop  
    alarmActive = false;  
  
    // --- Temperature Check ---  
    if (tempValue > TEMP_THRESHOLD) {  
        digitalWrite(LED_TEMP, HIGH);  
    }  
}
```

```

    alarmActive = true;
    Serial.println("!! TEMPERATURE DANGER DETECTED !!");
} else {
    digitalWrite(LED_TEMP, LOW);
}

// --- Gas Check ---
if (gasValue > GAS_THRESHOLD) {
    digitalWrite(LED_GAS, HIGH);
    alarmActive = true;
    Serial.println("!! GAS DANGER DETECTED !!");
} else {
    digitalWrite(LED_GAS, LOW);
}

// --- Buzzer Control ---
if (alarmActive) {
    digitalWrite(BUZZER, HIGH);
} else {
    digitalWrite(BUZZER, LOW);
}

// --- Reset Button Check ---
if (digitalRead(RESET_BTN) == LOW) {
    alarmActive = false;
    digitalWrite(BUZZER, LOW);
    digitalWrite(LED_TEMP, LOW);
    digitalWrite(LED_GAS, LOW);
    Serial.println(">> Alarm Reset by Operator.");
    delay(500);
}

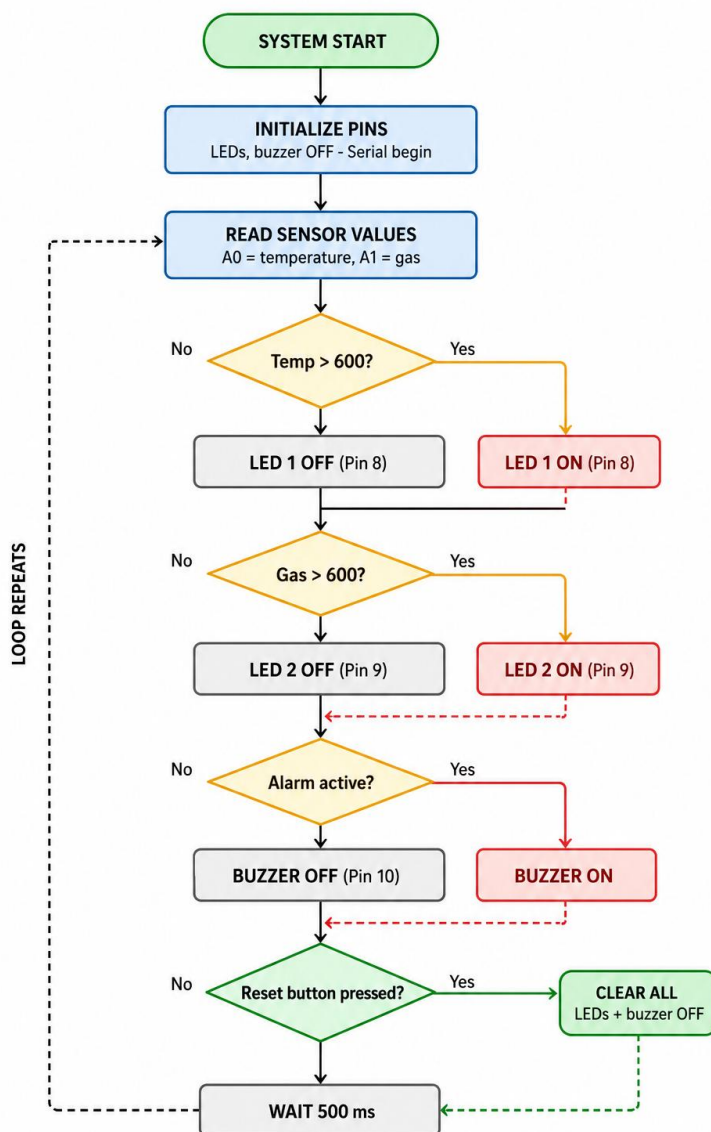
delay(500);
}

```

8. System Working

The system operates in a continuous monitoring loop. Below is a step-by-step explanation of how the system works:

- The Arduino reads analog values from both potentiometers every 500 milliseconds.
- If temperature sensor value (A0) exceeds 600, LED 1 (Pin 8) turns ON and alarm activates.
- If gas sensor value (A1) exceeds 600, LED 2 (Pin 9) turns ON and alarm activates.
- When any danger is detected, the buzzer (Pin 10) sounds continuously.
- When the reset button (Pin 12) is pressed, all alarms are cleared immediately.
- Sensor readings and events are printed to Serial Monitor for real-time tracking.



- **Start** → Initialize all pins
- **Read** sensor values every loop
- **Decision** diamonds for temperature and gas checks
- **LEDs and buzzer** turn ON/OFF based on readings
- **Reset button** clears everything
- **Loop** repeats every 500ms continuously

9. Simulation Test Results

All test scenarios were performed in SimulIDE and produced the expected results:

#	Test Scenario	Action Performed	Expected Result	Status
1	Normal Operation	Both potentiometers at low value	LEDs OFF, Buzzer silent	✓ Pass
2	Temperature Danger	Potentiometer 1 turned high	LED 1 ON, Buzzer ON	✓ Pass
3	Gas Danger	Potentiometer 2 turned high	LED 2 ON, Buzzer ON	✓ Pass
4	Both Dangers	Both potentiometers turned high	Both LEDs ON, Buzzer ON	✓ Pass
5	Alarm Reset	Push button pressed	All LEDs OFF, Buzzer OFF	✓ Pass

10. Conclusion

The Industrial Safety Monitoring and Control System was successfully designed, programmed, and simulated using Arduino UNO and SimulIDE. The system effectively demonstrates the core principles of embedded safety systems including real-time sensor monitoring, threshold-based alarm triggering, visual and audio alerts, and operator-controlled reset functionality.

This project lays a strong foundation for a real-world implementation. Future enhancements can include physical sensors (LM35, MQ-2, flame sensor), SMS/email alerts, data logging to SD card, and IoT-based remote monitoring for complete industrial safety coverage.

11. Project File Locations

File Type	File Name	Saved Location
Arduino Source Code	Industrial_Safety_System.ino	D:\Projects\ArduinoProjects\
Compiled HEX File	Industrial_Safety_System.ino.hex	D:\Projects\HEX_Files\
SimulIDE Circuit File	IndustrialSafety.sim1	D:\Projects\SimulIDE_Projects\